[4, 2, 1, 2, 5]

[1, 2, 2, 4, 5]

T = ~~8~~ ~~7~~ ~~5~~ 3

res = ~~0~~ 3.

TC = O(n log n)
AS = O(1).

W = 50   n = 3

vals: [60, 100, 120].
wts: [10, 20, 30]

W = ~~50~~ ~~40~~ ~~20~~ 0   W = ~~50~~ ~~30~~ 0

res = ~~0~~ ~~60~~ ~~160~~ 240.   res: ~~0~~ ~~120~~ 220.

vals: [80 100 120]
wts: [30 20 50]

W = 50.

res = 180.

Greedily
Considering item with max value first X.

vals: 10 20 100
wts: 10 40 50
W = 50

Greedily considering the items with smallest wt first → X

vals: [60 100 120]
wts: [10 20 30].
val/wt: 6 5 4

W = 50.

$\frac{6}{1}$ $\frac{6}{2}$ ------- $\frac{6}{10}$ $\frac{5}{11}$ ------- $\frac{5}{30}$ ... $\frac{4}{4}$ $\frac{4}{50}$

$60 + 100 + \frac{2}{3} \times \overset{40}{\cancel{120}} = 240.$

TC = O(n log n + n) = O(n log n)
AS = O(n).

[6, 5, 1]
10 20 30

W = ~~50~~ ~~40~~ ~~20~~ 0
res = ~~0~~ ~~60~~ ~~160~~ 240.

$\frac{v}{w} \times w$

{$\frac{v}{w}$, w}

V = [60 100 120]
W = [10 20 50].
6 5 2.4

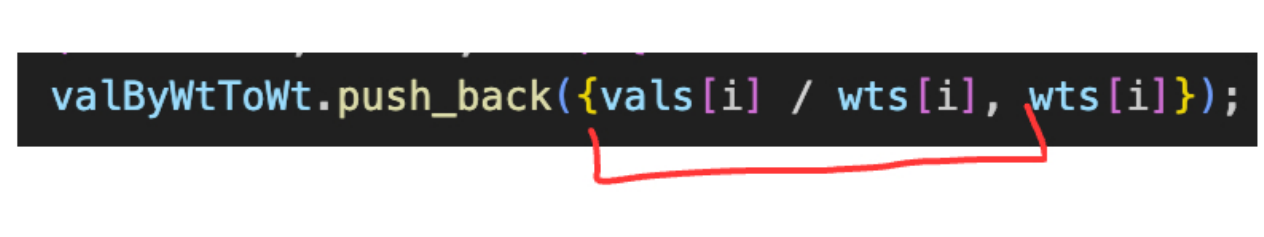
W = ~~50~~ ~~40~~ ~~20~~ 0.

res: ~~60~~ 160.

$160 + \frac{2}{5} \cdot 120 = 208.$

$\frac{12}{5}$

$\frac{240}{5}$



```
valByWtToWt.push_back({vals[i] / wts[i], wts[i]});
```

vals[i] = 120
wts[i] = 50

$\left(\frac{int}{int} \rightarrow int\right)$

$\frac{\rightarrow double}{int} \rightarrow double$

$\frac{int}{\rightarrow double} \rightarrow double.$

---

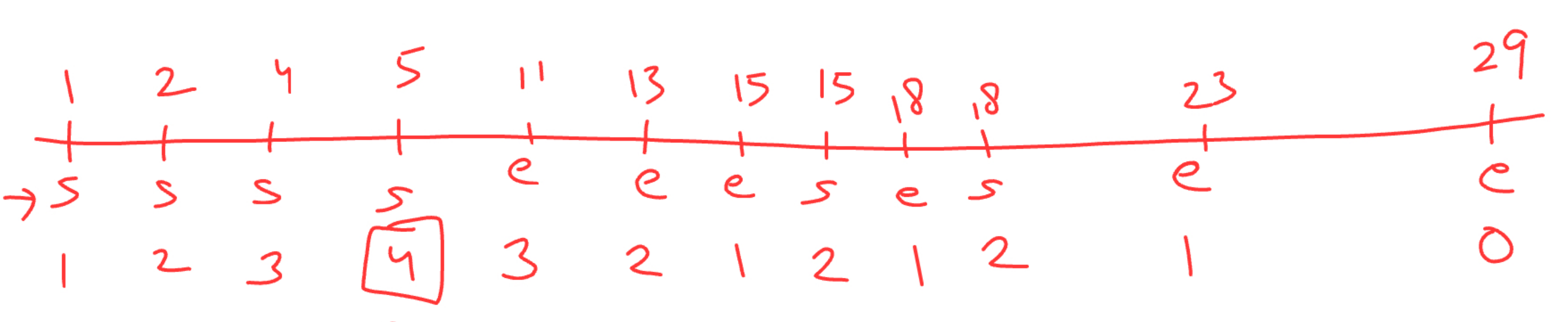
1 18 → 1
18 23 → 1
15 29 → 2
4 15 → 2
2 11 → 3
5 13 → 4.

1 15 → 1
15 20

1 18 → 1
2 11 → 2
4 15 → 3
5 13 → 4
15 29 → 4
18 23 → 5

1 10 → 1
→ 2 12 → 2
11 20 → ~~3~~ 1

No. of meeting rooms = No. of max. concurrent meetings taking place at any given point of time.



res = 4.

TC = O(n log n).
AS = O(n).

{int, char}
timestamp
's' / 'e'

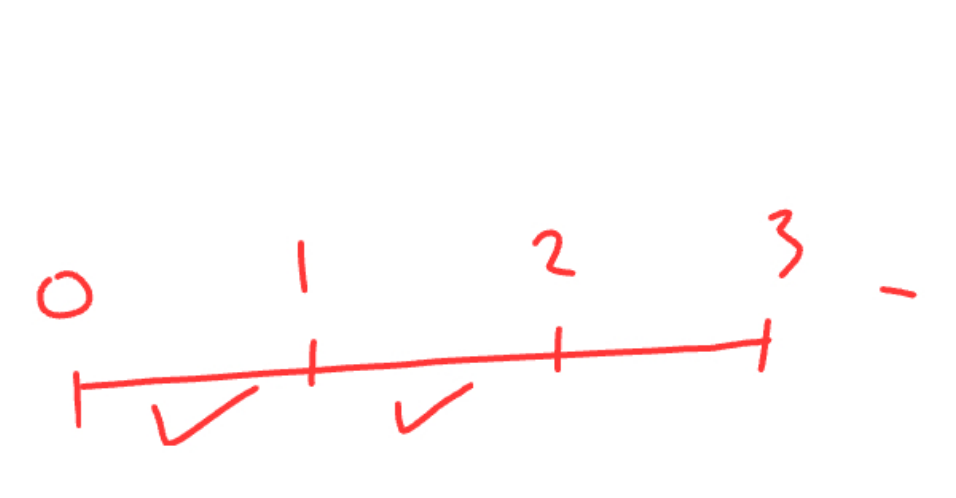
(10, s) (10, e).

---

id	deadline	profit.
→ 1	4	20 ←
2	1	10 ~~←~~
→ 3	1	40 ~~←~~
4	1	30 ~~←~~

max Profit = 60.
num Jobs = 2

10

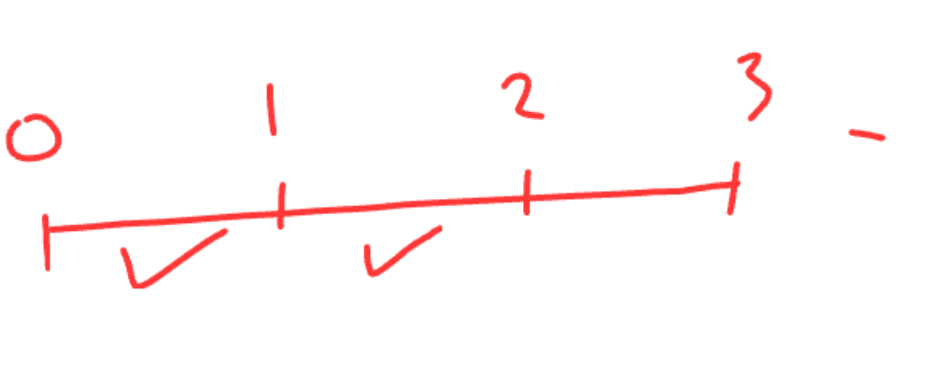


Profit = ~~40~~ 60
num = ~~1~~ 2

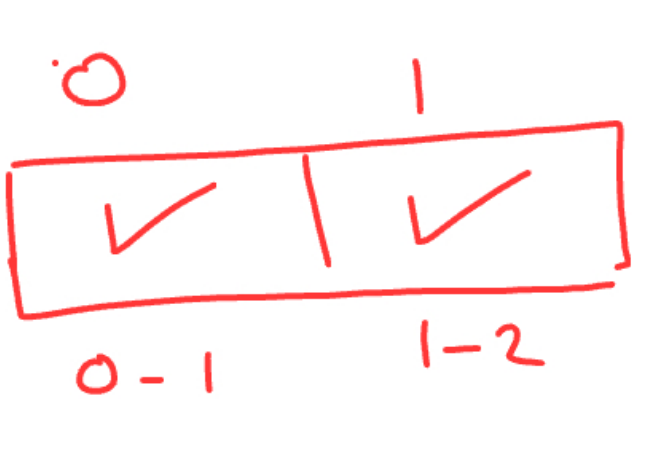
1	2	100 ~~←~~
2	1	50 ←

Profit = 150.
num = 2.

P = ~~0~~ ~~100~~ 150
n = ~~0~~ ~~1~~ 2



slots:



i: i → (i+1)th slot.

TC: O(n log n + n* max_deadline).
AS:

P = 100 , 100
d = 5 , 1

1	2	100 ~~←~~
2	1	19 ~~←~~
3	2	27 ←
4	1	25 ~~←~~
5	1	15. ~~←~~

P = ~~25~~ 27 + 100
num = ~~1~~ 2